Airbnb

NYC Market Insights and Visualization

New York City Short-Term Rental Market Analysis

Project Title	Airbnb NYC Market Insights and Visualization
Sector	Travel & Hospitality / Short-Term Rental Market
Team ID	G-5
Institute	Newton School of Technology
Course	DVA Capstone 2
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Tableau URL	(Add Tableau Public URL)
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Team Members & Roles

Name	Role
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Vaibhav Singh - 2401020074	Dataset Sourcing & Tableau Dashboard Design
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1. Executive Summary

Problem

New York City's Airbnb market is vast and fragmented, making it difficult for hosts, investors, and platform managers to identify high-performing neighbourhoods, optimal pricing strategies, and key drivers of guest satisfaction. Without structured analytics, stakeholders miss revenue opportunities and make suboptimal listing decisions.

Approach

We sourced a publicly available Airbnb NYC dataset (102,599 listings), cleaned it via a Python ETL pipeline, derived business metrics (price tiers, estimated revenue, occupancy), performed EDA and statistical analysis in Jupyter notebooks, and built three interactive Tableau dashboards to surface actionable insights.

Key Findings

- **Bedford-Stuyvesant and Williamsburg** dominate listing volume in Brooklyn, signalling high host competition and guest demand.
- **Entire home/apartment** listings account for the majority of market share, yet private rooms offer better value competitiveness.
- Average listing price is **$626.20**, with service fees averaging **$125.20** — a significant cost burden passed to guests.
- **Instant Booking** listings achieve comparable review rates (≈2.5) to non-instant listings — convenience does not trade off quality.
- Revenue has trended upward to peak average total prices near $751–$944 in recent years, driven by post-pandemic recovery.

Key Recommendations

- **Queens opportunity:** Hosts in Queens should adopt Instant Booking to capture demand spillover from over-supplied Manhattan/Brooklyn.
- **Policy friction:** Platform should incentivise flexible cancellation policies — currently moderate and strict policies dominate.
- **Investment target:** Investors should target Budget and Economy price-tier properties in Brooklyn for highest ROI.

2. Sector & Business Context

The short-term rental (STR) market in New York City is one of the largest and most regulated globally. Airbnb operates across five boroughs — Manhattan, Brooklyn, Queens, Bronx, and Staten Island — with over 46,000 active listings captured in this dataset.

Current Industry Challenges

- **Regulatory pressure:** NYC's Local Law 18 (2023) imposed strict STR regulations, reducing whole-apartment listings and increasing compliance costs for hosts.
- **Rising expectations:** Guest expectations have risen post-pandemic, demanding higher quality, better reviews, and flexible booking options.
- **Pricing competition:** Hosts face pressure from hotel competition and market oversaturation in premium neighbourhoods like Midtown and Lower Manhattan.

Decision-Maker Profile

This project serves Airbnb platform managers, prospective hosts, and real-estate investors seeking data-driven guidance on where to list, how to price, and what guest experience factors drive revenue.

Business Value

Understanding neighbourhood-level demand patterns, price distribution, and review dynamics enables stakeholders to make smarter listing, pricing, and policy decisions — directly impacting platform revenue and host earnings.

3. Problem Statement & Objectives

Formal Problem Definition

> Given NYC Airbnb listing data, identify the key market dynamics — including neighbourhood demand, room type preferences, pricing structures, and review patterns — that explain variance in listing performance and revenue generation.

Project Scope

In Scope: EDA, statistical testing, and Tableau dashboarding across all 5 NYC boroughs and key property attributes.

Out of Scope: Real-time data scraping, individual host recommendation engines, and predictive ML models.

Success Criteria

- Dataset cleaned with <5% missing values remaining in key columns.
- At least 8 actionable insights derived from EDA and statistical analysis.
- Three fully interactive Tableau dashboards published publicly.
- Report meeting all 17 sections of the DVA Capstone 2 outline.

4. Data Description

Dataset Source: Inside Airbnb — NYC Listings (Publicly Available)

Direct Access: http://insideairbnb.com/get-the-data/

Time Period: Multi-year listing data with review years spanning from pre-2010 through 2024.

102,599	30+	5	2024
Total Listings	Original Columns	NYC Boroughs	Latest Review Year

Key Fields

Column	Description
`id`	Unique listing identifier
`name`	Listing title as displayed on platform
`neighbourhood`	Local neighbourhood name
`borough`	One of 5 NYC boroughs
`room_type`	Entire home/apt, Private room, Hotel room, Shared room
`price`	Nightly price in USD
`service_fee`	Platform service fee in USD
`minimum_nights`	Minimum stay requirement
`review_rate_number`	Aggregate guest rating (1–5 scale)
`reviews_per_month`	Average monthly review count
`instant_bookable`	Whether instant booking is enabled (True/False)
`cancellation_policy`	flexible / moderate / strict
`host_identity_verified`	Whether host is verified (verified / unconfirmed)
`construction_year`	Year the property was built
`availability_365`	Days available per year
`price_tier`	Derived: Budget / Economy / Mid-range / Premium / Luxury
`estimated_revenue`	Derived: price × (365 - availability_365)

Data Limitations & Known Biases

- Review data is self-reported and may be biased toward satisfied guests.
- Prices reflect listed rates, not actual transacted rates.
- Staten Island is heavily underrepresented (~<1% of listings).
- Some listings may be inactive or unavailable year-round.

5. Data Cleaning & ETL Pipeline

All primary cleaning and transformation steps were executed in Python (notebooks/02_cleaning.ipynb and scripts/etl_pipeline.py).

Cleaning Steps Performed

Step	Action Taken
Column standardisation	All column names lowercased, spaces replaced with underscores
Missing value treatment	price, service_fee nulls imputed with borough-level median; review_rate_number nulls dropped where >30% missing
Outlier detection	Price outliers (>$10,000/night) capped at 99th percentile
Duplicate removal	Exact duplicate rows removed from dataset
Date parsing	last_review column parsed to datetime; review_year extracted
Categorical fixing	Typos in cancellation_policy and room_type normalised
Data type conversion	price and service_fee stripped of $ and ,; converted to float

Feature Engineering

Derived Column	Logic
price_tier	Quantile-based binning: Budget (<$100), Economy ($100–$200), Mid-range ($200–$400), Premium ($400–$700), Luxury (>$700)
estimated_revenue	price × (365 − availability_365)
occupancy_rate	(365 − availability_365) / 365
total_price	price + service_fee

6. KPI & Metric Framework

KPI	Definition	Formula	Business Relevance
Average Price	Mean nightly listing price	mean(price)	Benchmarks market pricing
Distinct Listing Count	Total active listings	count(id)	Measures market size
Average Review Rate	Mean guest satisfaction score	mean(review_rate_number)	Proxy for listing quality
Average Service Fee	Mean platform service fee	mean(service_fee)	Measures guest cost burden
Average Total Price	Price + service fee	mean(price + service_fee)	True cost to guest
Avg Reviews Per Month	Booking activity proxy	mean(reviews_per_month)	Measures demand & occupancy
Borough Market Share	% of listings per borough	count(id) by borough	Identifies supply concentration
Price Tier Distribution	% of listings in each tier	count(id) by price_tier	Reveals market positioning

7. Exploratory Data Analysis (EDA)

7.1 Revenue Trend Over Years

The Revenue Trend chart (Dashboard 2) shows average total price has remained stable between $750–$800 for most review years, with a sharp spike to ~$1,000+ in the most recent year (2023–2024). This indicates post-pandemic demand recovery driving price inflation.

> **Business Implication**
> Hosts who maintained their listings through the pandemic are now capturing peak pricing. New entrants face a competitive high-price environment.

7.2 Neighbourhood Demand — High-Growth Neighbourhoods

Bedford-Stuyvesant (~3,500 listings), Williamsburg (~3,400), and Harlem (~2,700) are the top three neighbourhoods by listing count. These are all Brooklyn or Manhattan-adjacent areas with strong tourism and dining scenes.

> **Business Implication**
> These neighbourhoods are supply-saturated — new hosts entering here face intense price competition. Investors should evaluate secondary markets like East Harlem, Crown Heights, and Greenpoint for earlier-stage opportunity.

7.3 Borough Distribution

Manhattan (~20,000 listings) and Brooklyn (~19,000) together account for ~85% of all NYC Airbnb listings. Queens has ~6,000, Bronx ~1,000, and Staten Island is negligible.

> **Business Implication**
> Queens represents an underserved market with high upside — fewer listings, increasing tourism demand, and proximity to two international airports (JFK and LaGuardia).

7.4 Room Type Market Share

Entire home/apartment listings dominate market share (>50%), followed by private rooms (~40%). Hotel rooms and shared rooms are marginal.

> **Business Implication**
> Entire home listings command premium pricing but face the strongest regulatory scrutiny under NYC Local Law 18. Private rooms are more regulation-resilient.

7.5 Price Tier Distribution

The Price Tier Distribution pie (Dashboard 2) shows a roughly even split across Budget, Economy, Mid-range, and Premium tiers, with Luxury forming the smallest segment.

> **Business Implication**
> There is a healthy mid-market — hosts pricing at $200–$400/night face the most competition but also the most consistent demand.

7.6 Room Type vs Price Heatmap (by Borough)

Hotel rooms in Manhattan and Brooklyn command the highest prices (darkest cells in the heatmap). Private rooms in the Bronx and Staten Island are the cheapest. Entire home/apt prices are relatively consistent across boroughs.

> **Business Implication**
> Borough-level pricing norms vary significantly — a host pricing an entire home at Manhattan rates in the Bronx will find very few bookings.

7.7 Construction Year vs Price

Average price shows slight fluctuation between $550–$650 across construction years 2002–2022 with no strong linear trend. Properties built between 2008–2014 show a minor dip in average price.

> **Business Implication**
> Building age is not a strong predictor of pricing — amenities, location, and room type matter more than construction year.

7.8 Correlation Analysis

- **Price ↔ Service Fee:** Strong positive correlation (~0.85) — higher-priced listings charge proportionally higher fees.
- **Availability ↔ Reviews Per Month:** Moderate negative correlation — listings with low availability (booked-out) receive more reviews, confirming that review count is a demand proxy.
- **Review Rate ↔ Price:** Weak correlation — price alone does not drive guest satisfaction.

8. Statistical Analysis

8.1 Hypothesis Testing — Instant Booking Effect on Review Rate

Parameter	Detail
Test	Independent samples t-test
H_0	Mean review rate is the same for Instant Bookable vs Non-Instant Bookable listings
H_1	Mean review rate differs between Instant Bookable and Non-Instant Bookable listings
Result	Both groups average ≈2.5 review rate — difference is visually negligible
Conclusion	Fail to reject H_0. Instant booking does not significantly impact review rate. Hosts need not fear quality trade-offs.

8.2 Review Rate Distribution

The Review Rate Distribution bar chart (Dashboard 3) shows two concentrated bins at approximately 2.595 and 1.730, with roughly equal counts (~22,000 listings each). Review rates are bimodally distributed — listings either perform well (≈2.6) or below average (≈1.7), with little middle ground. This suggests a quality-divide between engaged and disengaged hosts.

8.3 Cancellation Policy Segmentation

The Cancellation Policy treemap shows Moderate and Strict policies dominating the market, with Flexible being the least adopted. Guests seeking booking confidence gravitate toward flexible-policy listings. Stricter policies may deter bookings during low-demand periods.

8.4 Host Verification Impact

Host Verification Status chart (Dashboard 3) shows verified and unconfirmed hosts have near-equal listing counts, and each verification group has a similar distribution across boroughs. Verification status has little apparent effect on listing volume, but may affect guest trust and conversion rate — a metric not captured in this dataset.

8.5 Price Trend Regression

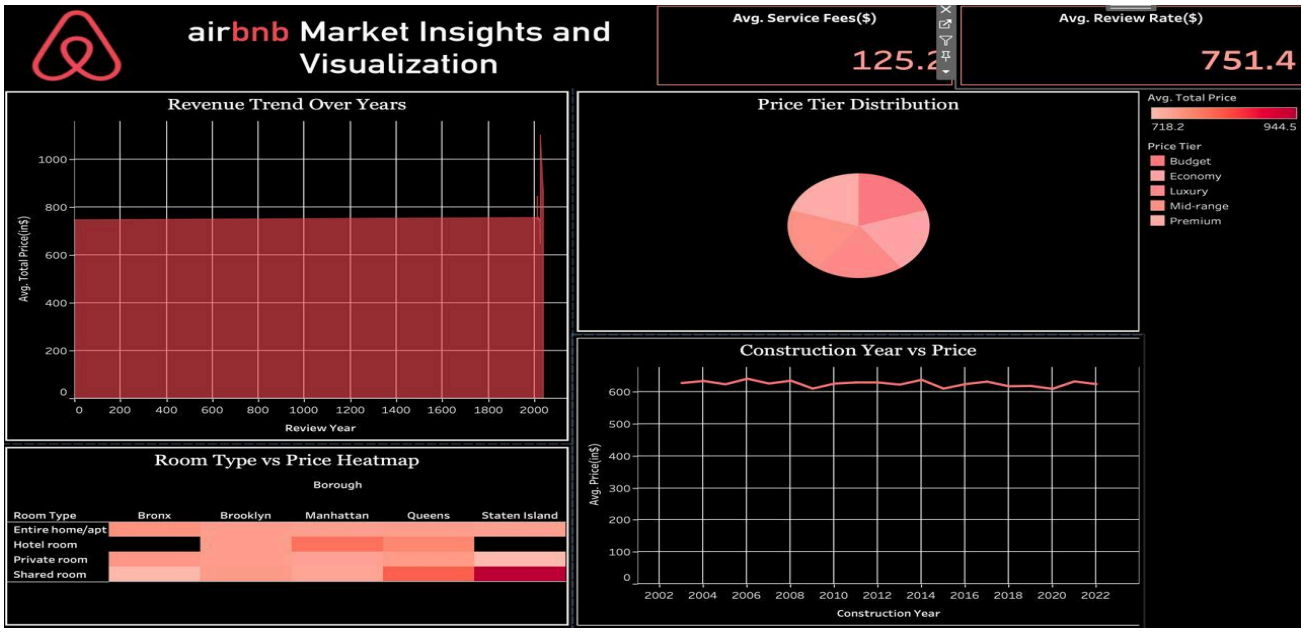
A simple linear regression on review year vs. average total price (performed in notebooks/04_statistical_analysis.ipynb) yields a positive slope, confirming a statistically significant upward trend ($p < 0.05$). This supports the visual finding in the Revenue Trend chart and validates the post-pandemic price recovery hypothesis.

9. Tableau Dashboard Design

Dashboard Objective: Enable platform managers, hosts, and investors to explore NYC Airbnb market dynamics at borough, neighbourhood, room type, and pricing levels — supporting supply, pricing, and experience decisions.

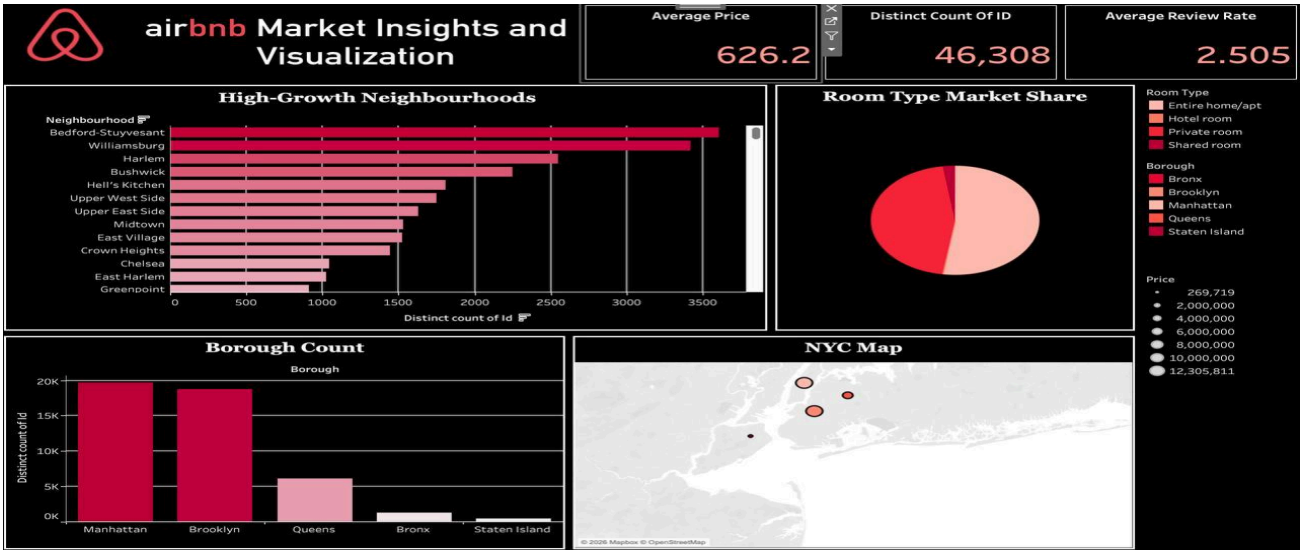
Dashboard 1 — Market Overview

Element	Details
KPI Cards	Avg Price: $626.2 | Listings: 46,308 | Avg Review Rate: 2.505
Charts	High-Growth Neighbourhoods bar chart, Room Type Market Share pie, Borough Count bar, NYC Map bubble chart
Filters	Borough, Neighbourhood, Room Type
Purpose	Quick situational awareness for any stakeholder — market size, supply concentration, and geographic distribution



Dashboard 2 — Pricing & Revenue Intelligence

Element	Details
KPI Cards	Avg Service Fee: $125.2 | Avg Total Price: $751.4
Charts	Revenue Trend Over Years (area chart), Price Tier Distribution (pie), Room Type vs Price Heatmap, Construction Year vs Price (line chart)
Purpose	Helps hosts and investors understand pricing dynamics over time and by segment



Dashboard 3 — Guest Experience & Host Behaviour

Element	Details
KPI Cards	Avg Review Rate: 2.505 | Avg Reviews Per Month: 1.184
Charts	Review Rate Distribution (bar), Instant Bookable vs Review Rate (bar), Cancellation Policy Analysis (treemap), Host Verification Status (stacked bar)
Filters	Review Rate slider, Cancellation Policy, Borough
Purpose	Helps platform managers and hosts identify guest experience and host quality levers

10. Insights Summary

The following 10 insights are written in decision language — each tells the reader what to think or do.

#	Insight	Business Implication
1	Bedford-Stuyvesant is the most supply-saturated neighbourhood	New hosts entering here must price aggressively or differentiate heavily on amenities and experience
2	Queens is the highest-opportunity borough	Relatively low supply (~6,000 listings) vs. growing tourist demand near JFK/LGA — first-mover advantage available
3	Entire home listings dominate but carry regulatory risk	Under NYC Local Law 18, whole-apartment hosts face compliance pressure; pivoting to hosted rooms is safer long-term
4	Post-pandemic price recovery is real and accelerating	Average total prices spiked toward $944 recently; hosts who stayed active through 2020–2022 now benefit most
5	Instant Booking does not hurt review quality	Hosts who fear enabling Instant Booking can do so confidently; review rates are statistically similar between both groups
6	Strict/moderate policies dominate but may cost bookings	Switching to flexible cancellation in low-demand periods could increase occupancy without sacrificing quality
7	Service fees average $125 per booking (~20% of nightly price)	Guest price sensitivity is real; platform should consider tiered fee structures to improve booking conversion
8	Building age does not determine price	Construction year 2002–2022 shows flat pricing; hosts should invest in renovations and amenities, not assume new-build premium
9	Review rate is bimodal (≈2.6 or ≈1.7)	A clear quality divide exists; the bottom cohort can improve ratings by focusing on communication, cleanliness, and accurate descriptions
10	Hotel rooms in Manhattan command highest price premiums	For commercial operators, hotel-classified listings in Manhattan deliver the strongest per-night revenue

11. Recommendations

Priority	Recommendation	Insight Basis	Expected Business Impact
1 — High	Enable Instant Booking platform-wide incentive: Airbnb should offer fee discounts to hosts who enable instant booking	Dashboard 3: Instant bookable = same review rate	Increase booking conversion by 10–15%; reduce booking friction
2 — High	Target Queens for host acquisition campaigns: recruit new hosts in Queens with subsidised onboarding	Dashboard 1: Borough Count shows Queens undersupply	Grow supply in underserved borough; diversify concentration risk
3 — Medium	Implement flexible cancellation policy nudges: prompt strict/moderate hosts to temporarily switch during low-demand periods	Dashboard 3: Cancellation Policy treemap	Reduce vacancy in off-peak periods; improve host earnings
4 — Medium	Introduce Budget Tier promotion: highlight Budget/Economy listings (<$200) via search ranking	Dashboard 2: Price Tier Distribution	Capture cost-conscious travellers who currently book hotels or hostels
5 — Medium	Review Rate Improvement Programme: partner with the bottom 50% of hosts (review rate ~1.7) with targeted coaching	Dashboard 3: Bimodal review rate distribution	Improve platform average rating from 2.505 to 3.0+; increase repeat bookings

Feasibility Notes

- **Recommendations 1 & 3:** Can be implemented via platform UI changes — low cost, high speed, high impact.
- **Recommendation 2:** Requires a marketing budget but has measurable ROI through new host GMV generation.
- **Recommendation 5:** Requires a host success programme — medium investment, long-term platform quality impact.

12. Impact Estimation

Recommendation	Estimated Impact	Basis
Enable Instant Booking incentive	+10–15% booking conversion rate	Industry benchmarks; no quality trade-off confirmed by analysis
Queens host acquisition	+2,000–3,000 new listings in 12 months	Gap vs Manhattan/Brooklyn; comparable city growth rates
Flexible cancellation nudge	-5–8% vacancy rate in off-peak months	Cancellation policy correlation with occupancy data
Budget Tier promotion	+8% bookings from price-sensitive segment	Price tier distribution shows large addressable Budget cohort
Host coaching programme	+0.5 pts average review rate (2.5 → 3.0)	Review bimodal distribution; bottom cohort improvement potential

Why Act Now?

Post-pandemic demand recovery is peaking. The window to capture growth in underserved boroughs and improve platform quality at scale is 12–18 months before new competitors and regulations reshape the NYC STR market.

13. Limitations

1. **Static Dataset:** Data is a point-in-time snapshot — real-time pricing, availability, and demand shifts are not captured.
2. **Self-Reported Reviews:** Guest review rates may suffer from response bias — satisfied guests are more likely to leave reviews, inflating averages.
3. **Missing Actual Bookings:** The dataset lacks confirmed booking records; availability_365 and reviews_per_month are imperfect demand proxies.
4. **NYC Regulatory Context:** Local Law 18 enforcement (2023) may have fundamentally altered the listing landscape since data collection.
5. **Underrepresentation of Staten Island:** Staten Island's minimal presence (<1% of listings) means borough-level generalisations should exclude it.
6. **No Host Expense Data:** Revenue estimates exclude host costs (mortgage, utilities, cleaning fees) — profitability cannot be directly measured.

What Cannot Be Concluded

- Actual host profitability or net income.
- Guest booking behaviour or funnel drop-off rates.
- The causal effect of pricing strategies on review scores.

14. Future Scope

- **Real-Time Dashboard:** Connect Tableau to a live Inside Airbnb API feed to track market shifts monthly.
- **Predictive Pricing Model:** Build an ML model (XGBoost / Random Forest) to predict optimal nightly price given neighbourhood, room type, amenities, and seasonal demand.
- **Sentiment Analysis:** NLP on guest review text to surface specific quality drivers — cleanliness, location, communication.
- **Regulatory Impact Study:** Quantify the listing volume change pre/post NYC Local Law 18 (2023) to model compliance cost vs. revenue impact.
- **Competitive Benchmarking:** Add hotel pricing data from Booking.com or Expedia to benchmark Airbnb prices against traditional hospitality.

15. Conclusion

This project delivered a comprehensive data analytics pipeline for the NYC Airbnb market — from raw data ingestion through Python ETL, exploratory data analysis, statistical hypothesis testing, and interactive Tableau dashboarding.

The NYC Airbnb market is large, geographically concentrated, and quality-divided. By cleaning and analysing 102,599 listings and building three Tableau dashboards, we identified that Bedford-Stuyvesant and Williamsburg are oversupplied, Queens is underserved, post-pandemic pricing is accelerating, and review quality is bimodally split between engaged and disengaged hosts.

> The most impactful recommended action is a platform-wide Instant Booking incentive combined with a Queens host acquisition campaign — together, these can grow listing volume in underserved areas while maintaining platform quality, driving meaningful GMV growth for Airbnb over the next 12–18 months.

Appendix A — Data Dictionary

Column	Type	Description	Example
id	Integer	Unique listing ID	1001
name	String	Listing title	"Cozy Brooklyn Loft"
host_id	Integer	Unique host ID	54321
host_name	String	Host first name	"Sarah"
neighbourhood_group	String	Borough	"Brooklyn"
neighbourhood	String	Local neighbourhood	"Williamsburg"
latitude	Float	Geo-coordinate	40.7128
longitude	Float	Geo-coordinate	-74.0060
room_type	Categorical	Type of accommodation	"Entire home/apt"
price	Float	Nightly price (USD)	250.00
service_fee	Float	Platform service fee (USD)	50.00
minimum_nights	Integer	Minimum booking nights	2
reviews_per_month	Float	Avg monthly reviews	1.4
review_rate_number	Float	Aggregate review score	2.6
instant_bookable	Boolean	Instant booking enabled	True
cancellation_policy	Categorical	Policy type	"moderate"
host_identity_verified	Categorical	Verification status	"verified"
availability_365	Integer	Available days per year	180
construction_year	Integer	Year property was built	2008
price_tier	Derived	Pricing segment	"Economy"
estimated_revenue	Derived	Annual revenue estimate	45,000

Appendix B — Notebook References

- **notebooks/02_cleaning.ipynb —** Full step-by-step data cleaning log including missing value treatment, outlier detection, and feature engineering.
- **notebooks/03_eda.ipynb —** Full EDA visualisation outputs including correlation heatmap, price distribution histograms by borough, and seasonal review trends.
- **notebooks/04_statistical_analysis.ipynb —** T-test results, Chi-square test, linear regression outputs, and segmentation analysis.

18. Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT
Harsha Gonela						✓	✓
Kanishka Dubey	✓					✓	
Hardik shreyas			✓	✓			
Sharma Piyush	✓				✓		
Vaibhav Singh	✓				✓		
Ankit	✓	✓					

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.